



SURGIN: SURrogate-guided Generative INversion for Geological Carbon Sequestration

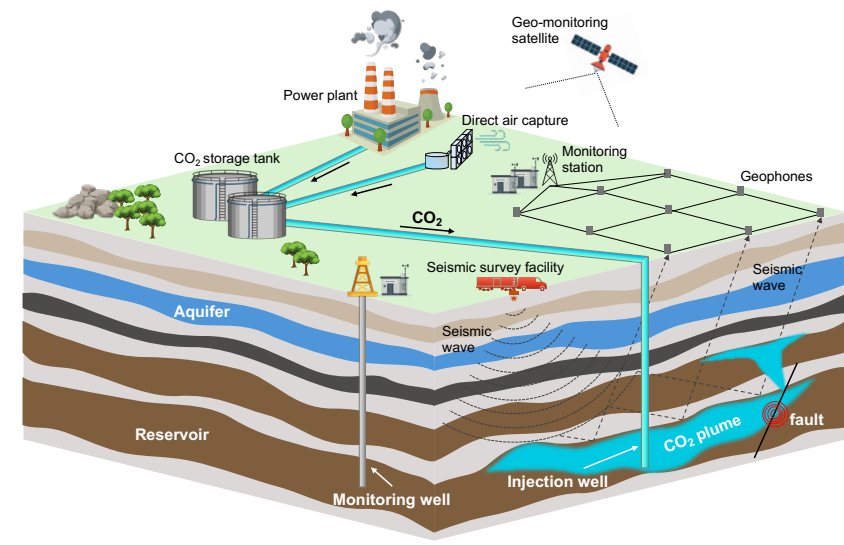
Zhao Feng¹, Bicheng Yan², Fengshou Zhang¹

¹ Department of Geotechnical Engineering, Tongji University, Shanghai, China

² Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia



1. Background



Bayesian inverse problem:

$$p(\mathbf{m}|\psi) = \frac{p(\mathbf{m})p(\psi|\mathbf{m})}{p(\psi)}$$

Stein score function form:

$$\begin{aligned} \nabla_{\mathbf{m}} \log p(\mathbf{m}|\psi) &= \nabla_{\mathbf{m}} \log p(\mathbf{m}) + \nabla_{\mathbf{m}} \log p(\psi|\mathbf{m}) \\ &\quad \text{Prior} \quad \text{Likelihood} \end{aligned}$$

$\mathbf{m}$: geo-parameter

ψ : observations

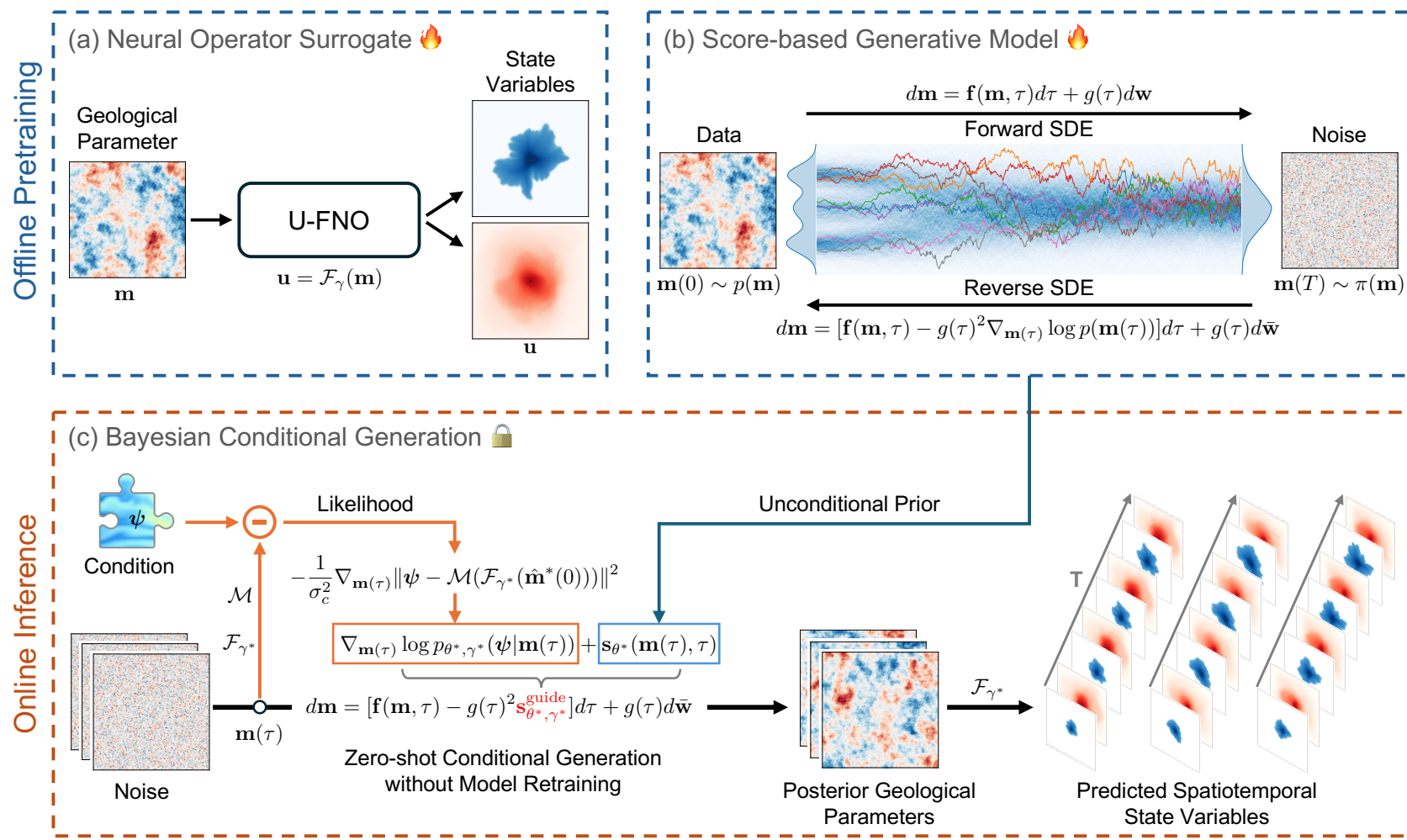
- The inherent uncertainty in geological parameters and scarcity of monitoring data present a fundamental challenge to the accurate characterization of state variables in geological carbon sequestration (GCS).
- Conventional inversion methods rely on numerical simulations to evaluate the likelihood function through iterative forward modeling. — **Computationally expensive**
- Machine learning (ML)-based surrogate models can replace high-fidelity simulations, thereby accelerating the inversion process. However, surrogate-based inversion necessitates re-running the entire workflow if observational data is changed. — **Lack generalizability**

The proposed SURGIN model:

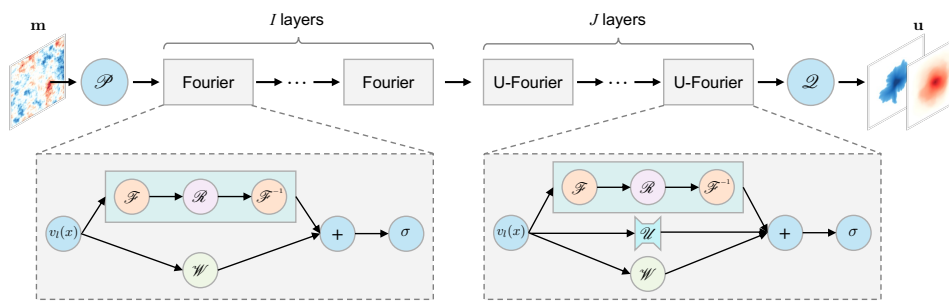
- ✓ integrates surrogate approximation, generative learning, and Bayesian inference
- ✓ can **zero-shot** generate posterior geological parameters based on unseen conditions
- ✓ enables real-time **forward** and **inverse** modeling in a unified generative framework

2. Methods

➤ Overview:



➤ U-FNO surrogate model:



Function space mapping:

$$\mathcal{F}_{\gamma} : \mathbf{m} \in \mathbb{R}^{N_m} \rightarrow \mathbf{u} \in \mathbb{R}^{N_u \times N_t}$$

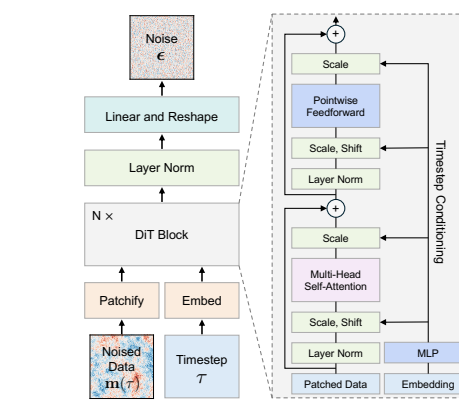
Regression loss function:

$$\gamma^* = \arg \min_{\gamma} \mathbb{E}_{p(\mathbf{m})} [\|\mathcal{F}_{\gamma}(\mathbf{m}) - \mathcal{F}(\mathbf{m})\|^2]$$

Surrogate approximation:

$$\mathbf{u} \simeq \mathcal{F}_{\gamma^*}(\mathbf{m})$$

➤ Score-based generative model:



Forward stochastic differential equation:

$$d\mathbf{m} = \mathbf{f}(\mathbf{m}, \tau)d\tau + g(\tau)d\mathbf{w}$$

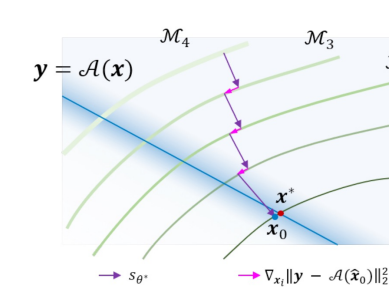
Reverse stochastic differential equation:

$$d\mathbf{m} = [\mathbf{f}(\mathbf{m}, \tau) - g(\tau)^2 \nabla_{\mathbf{m}(\tau)} \log p(\mathbf{m}(\tau))]d\tau + g(\tau)d\mathbf{w}$$

Score matching loss function:

$$\mathcal{L}^{\text{SM}}(\theta) = \mathbb{E}_{p(\tau)} \left[\lambda(\tau) \mathbb{E}_{p(\mathbf{m}(\tau))} [\|\mathbf{s}_{\theta}(\mathbf{m}(\tau), \tau) - \nabla_{\mathbf{m}(\tau)} \log p(\mathbf{m}(\tau))\|^2] \right]$$

➤ Bayesian conditional generation:



Relationship between observation and geological parameters:

$$\psi \simeq \mathcal{M}(\mathcal{F}_{\gamma^*}(\mathbf{m})) + \epsilon_c$$

The likelihood score is (by Tweedie's formula & Jensen's inequality):

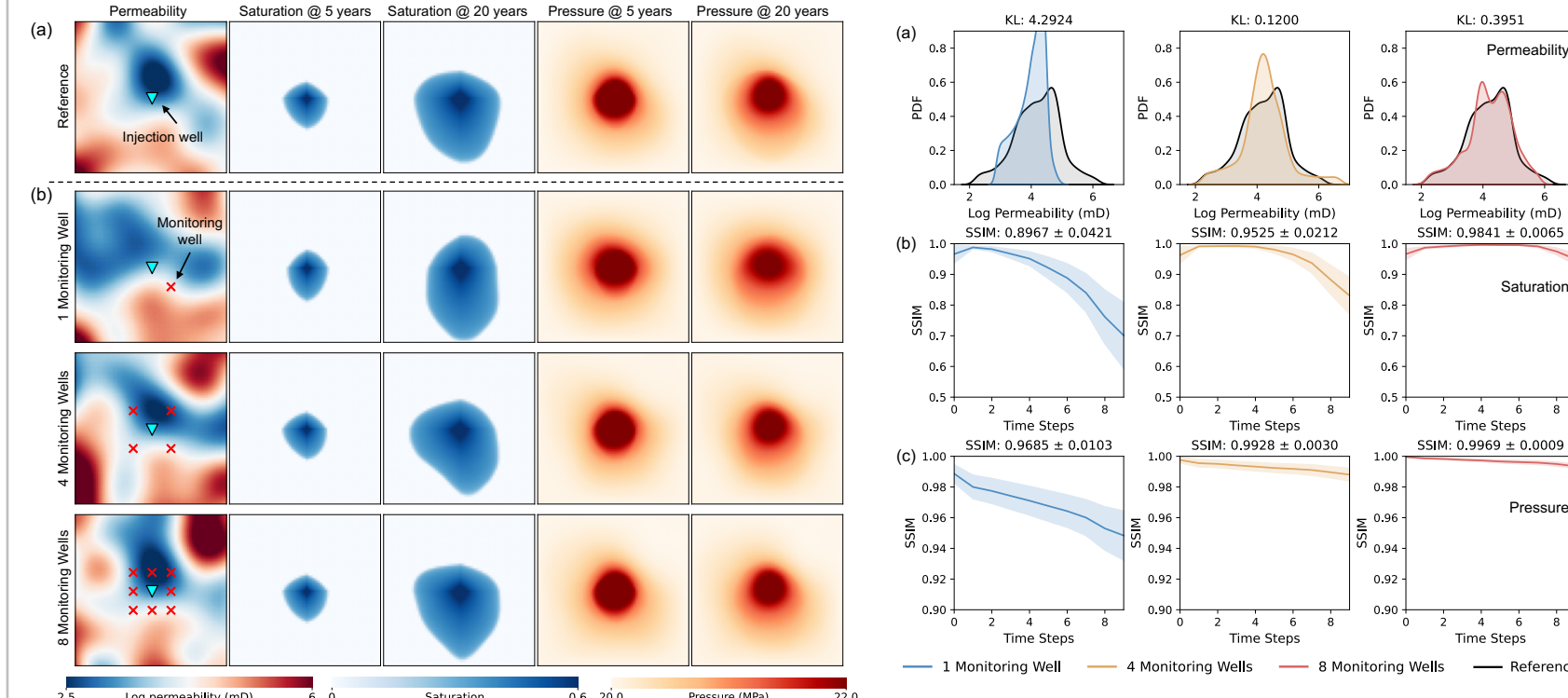
$$p(\psi|\mathbf{m}(\tau)) \simeq p(\psi|\hat{\mathbf{m}}(0))$$

Chain rule through automatic differentiation (AD):

$$\begin{aligned} \nabla_{\mathbf{m}(\tau)} \log p(\psi|\mathbf{m}(\tau)) &\simeq \nabla_{\mathbf{m}(\tau)} \log p_{\theta^*, \gamma^*}(\psi|\mathbf{m}(\tau)) \\ &= -\frac{2}{\sigma_c^2} \left(\psi - \mathcal{M}(\mathcal{F}_{\gamma^*}(\hat{\mathbf{m}}^*(0))) \right) \frac{\partial \mathcal{M}(\mathcal{F}_{\gamma^*})}{\partial \mathcal{F}_{\gamma^*}} \frac{\partial \mathcal{F}_{\gamma^*}(\hat{\mathbf{m}}^*; \gamma^*)}{\partial \hat{\mathbf{m}}^*} \frac{\partial \hat{\mathbf{m}}^*(\mathbf{m}(\tau); \theta^*)}{\partial \mathbf{m}(\tau)} \end{aligned}$$

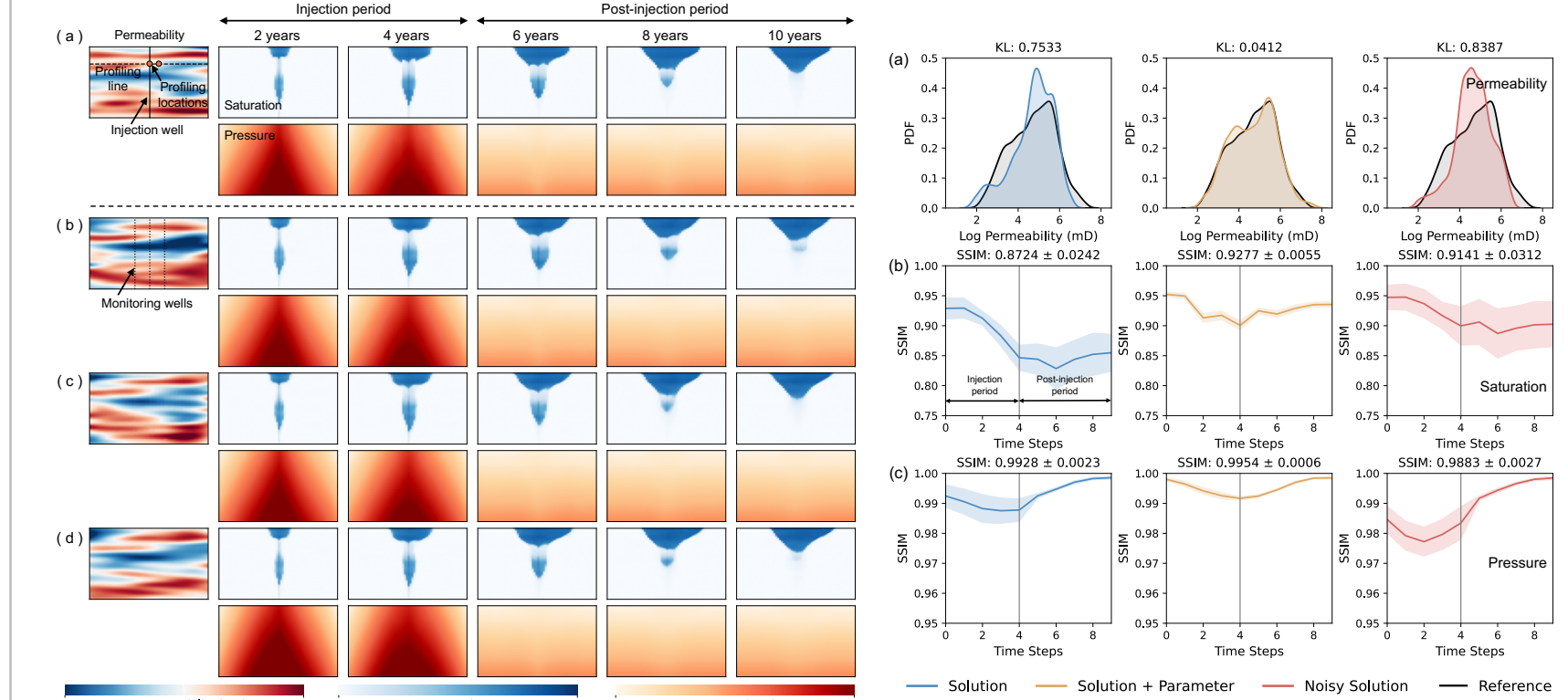
3. Results

➤ Sparse well measurements for the horizontal case:



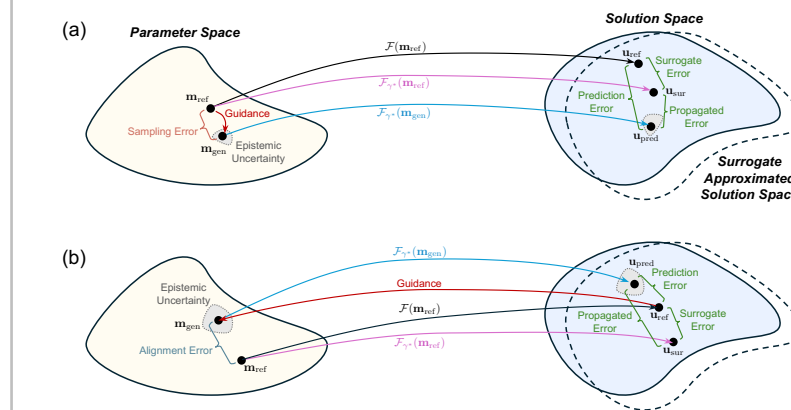
- SURGIN can reconstruct the full fields based on sparse well measurements
- The uncertainty decreases as more observation data are available

➤ History matching for the vertical case:



- SURGIN can infer the full fields based on incomplete and noisy monitoring data
- SURGIN's capability is not limited to the demonstrating cases. It can be used for various data assimilation tasks based on arbitrary observations **without model re-training**.

4. Discussion



Case	Conditioning scenario	Generation (sec/batch)	Simulation (min/sample)
1	Sparse well	45	~15
	Coarse parameters	14	
	Coarse solutions	28	
2	Sparse well	45	~10
	Corrupted parameters	14	
	Corrupted solutions	28	

- Aleatoric and epistemic uncertainties can be quantified within the framework
- SURGIN offers significant speedup against numerical simulation

Future work:

- Alignment error can be mitigated by introducing physics constraints during sampling
- High-dimensional unstructured data can be accommodated using dimension reduction

More information

- Acknowledgements: NSFC, Tencent CarbonX, CCF-Tencent Rhino-Bird Research, Tongji Lingjun Fellowship
- Code will be open sourced on: <https://github.com/fengzhao1239/DiffNO>
- See also our recent work: Conditional Neural Field Latent Diffusion for Geoscience (CoNFILd-geo) integrates neural field encoding with Bayesian diffusion model for data assimilation (arXiv:2508.16640)

